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Multi-Depot Metric TSP

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Problem

Input: Undirected graph $G = (V, E)$, metric weight $w\colon E \to \mathbb{N}$, set $D \subseteq V$ of d depots ■.

Task: Find a min-weight tour T: min-weight set of cycles (subtours) that visits all vertices, with a depot on every subtour.

3/2-Approximation in $2^{O(d \log d)}\, n^{O(1)}$ Time

Algorithm `Split`(G)

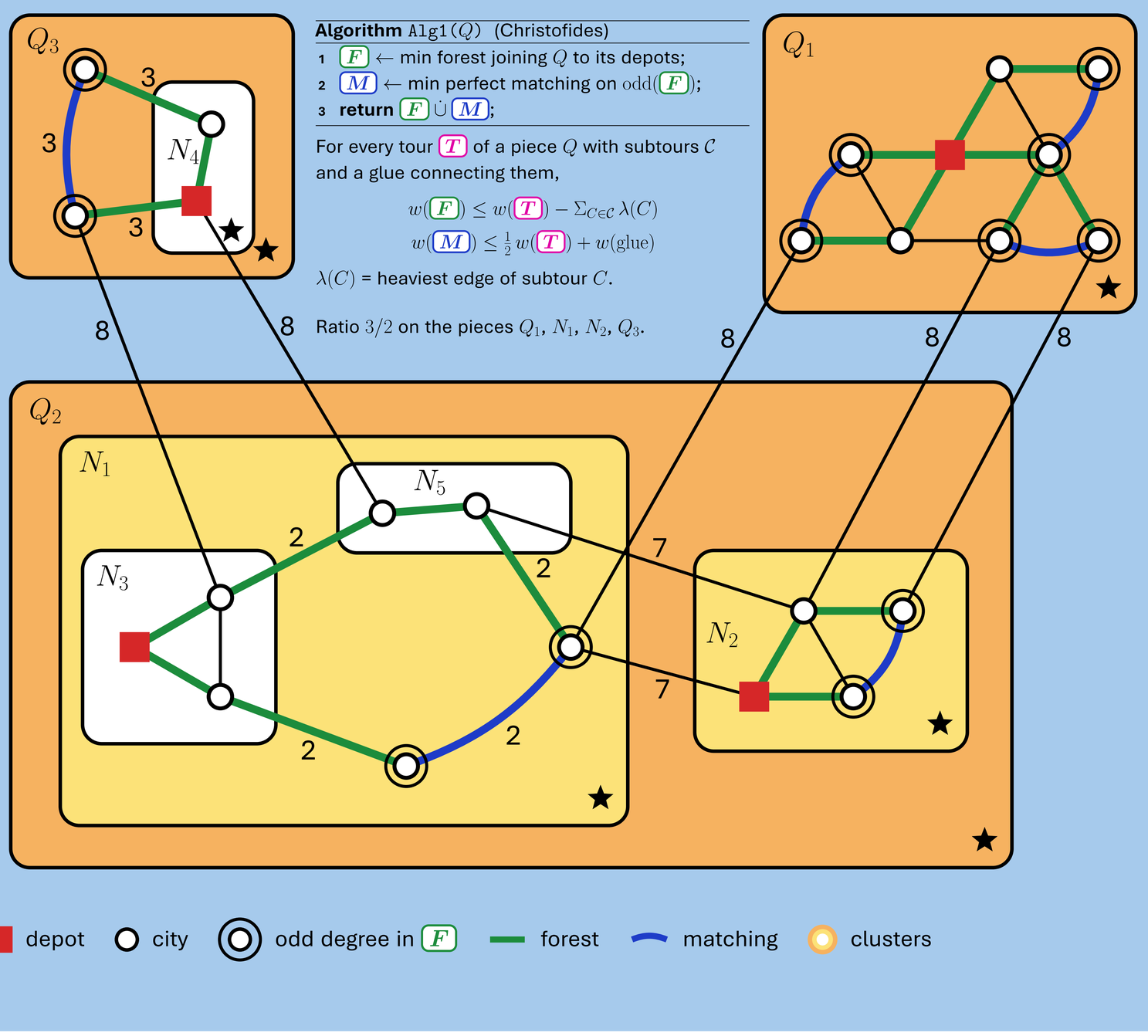
Input : Graph G, metric w, depots $D \subseteq V$.
Output : Tour T.

1 $F \leftarrow$ min-weight forest joining each vertex to D;
2 $M \leftarrow$ min-weight perfect matching on $\mathrm{odd}(F)$;
3 $T \leftarrow F \dot\cup M$;
} Alg 1
4 **foreach** depot s **do** $T \leftarrow \min\{T, \texttt{Split}(G \setminus s)\}$;
5 **foreach** candidate cluster N of G **do**
6 $\quad T \leftarrow \min\{T, \texttt{Split}(N) \dot\cup \texttt{Split}(G \setminus N)\}$;
7 **return** T;

Related Work

	ratio	time
Christofides, Serdyukov ($d = 1$)	$3/2$	n^3
Rathinam, Sengupta, Darbha '07	2	n^2
Xu, Xu, Rodrigues '11	$2 - 1/d$	n^3
Xu, Rodrigues '15	$3/2$	$n^{O(d)}$
Traub, Vygen, Zenklusen '20	$\alpha + \varepsilon$	$n^{f(d,\varepsilon)}$
Deppert, Kaul, Mnich '23 (rand.)	$3/2 + \varepsilon$	$(1/\varepsilon)^{O(d \log d)}\, n^3$
Zhao, Su, Xiao '26	$\alpha + \varepsilon$	$2^{O(d/\varepsilon)}\, n^3$
Zhao, Liu, Xiao '26	$1 + 1/\sqrt{2}$	n^3
This work	$\mathbf{3/2}$	$\mathbf{2^{O(d \log d)}\, n^3}$

Splitting on Candidate Clusters



Algorithm `Alg1`(Q) (Christofides)

1 $F \leftarrow$ min forest joining Q to its depots;
2 $M \leftarrow$ min perfect matching on $\mathrm{odd}(F)$;
3 **return** $F \dot\cup M$;

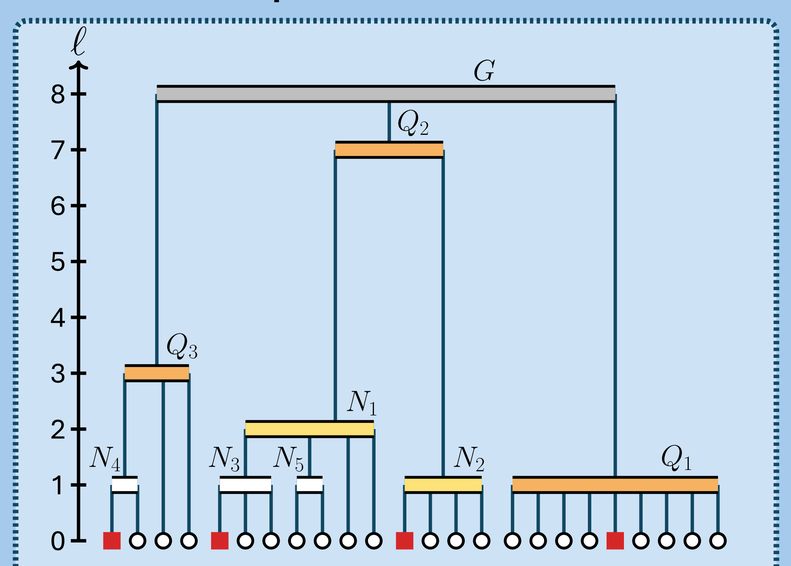
For every tour T of a piece Q with subtours $\mathcal{C}$ and a glue connecting them,

$$w(F) \le w(T) - \Sigma_{C \in \mathcal{C}}\, \lambda(C)$$
$$w(M) \le \tfrac{1}{2}\, w(T) + w(\mathrm{glue})$$

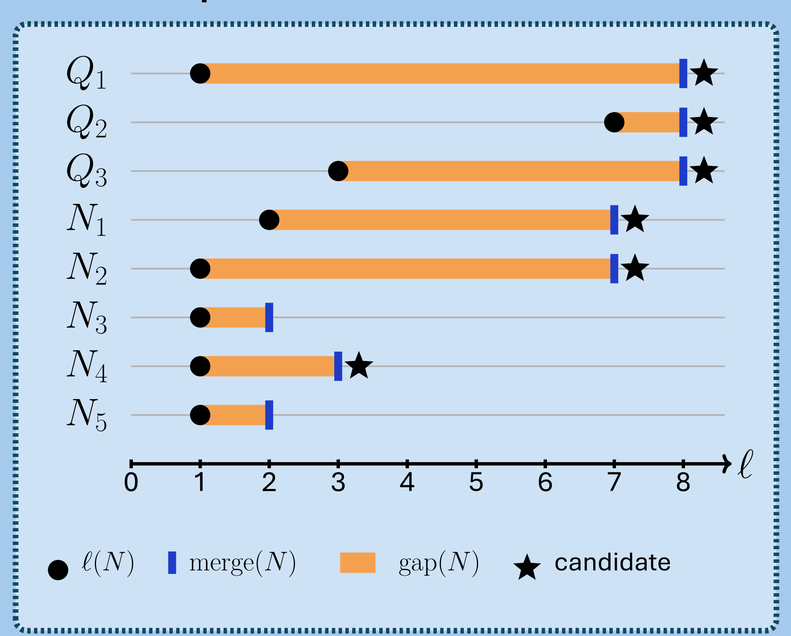
$\lambda(C)$ = heaviest edge of subtour C.

Ratio $3/2$ on the pieces Q_1, N_1, N_2, Q_3.

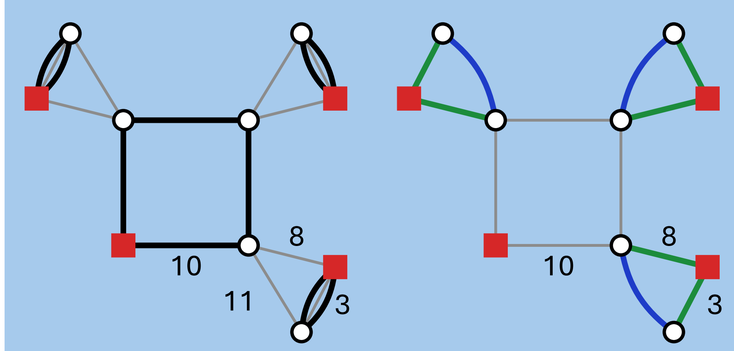
Cluster Decomposition



Cluster Gaps



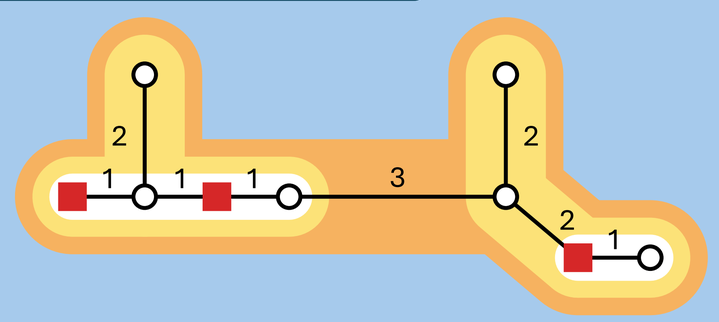
A Hard Case for Algorithm 1



Left: optimum, a ring and three loops, weight 58.
Right: Algorithm 1, weight 66.

Ring cities next to a loop are closer to its depot (8) than to the ring (10). The forest misses the ring, and the matching pays 11 per loop.

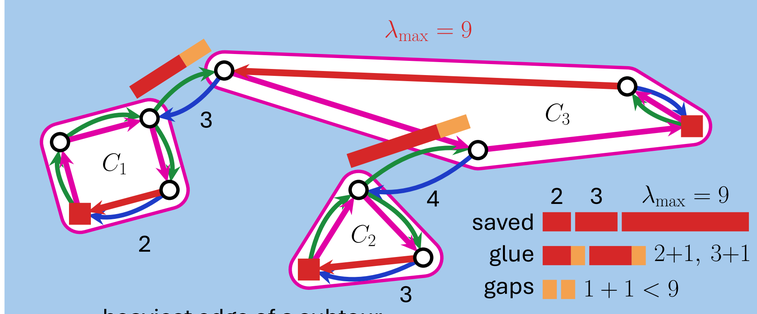
Growing Clusters



- **Cluster** N, a connected component after joining all pairs at distance $\le \ell$. $\boldsymbol{\ell(N)}$ = smallest such ℓ.
- $\mathbf{merge}(\boldsymbol{N})$ = smallest distance from N to the outside. $\mathbf{gap}(\boldsymbol{N}) = \mathrm{merge}(N) - \ell(N)$.
- **Candidate**, a cluster N with depots inside and outside and $(2d-2) \cdot \mathrm{gap}(N) \ge \mathrm{merge}(N')$, for N' the largest cluster with the same depots.

Offsetting Glued Subtours

Fix a tour T of a piece, with heaviest edge $\lambda_{\max}$.



- With one subtour, $w(F \dot\cup M) \le \frac{3}{2}\, w(T) - \lambda_{\max}$.
- $F \dot\cup M$ glues several subtours $\mathcal{C}$. Cluster N costs $\mathrm{merge}(N)$, a saved edge pays all but $\mathrm{gap}(N)$, $\lambda_{\max}$ stays.
  $w(\mathrm{glue}) - \Sigma_{C \in \mathcal{C}}\, \lambda(C) \le \Sigma_N\, \mathrm{gap}(N) - \lambda_{\max}$
- Non-candidates have $\mathrm{gap}(N) < \frac{\mathrm{merge}(N')}{2d-2} \le \frac{\lambda_{\max}}{2d-2}$, at most $2d-2$ of them, so $\Sigma_N\, \mathrm{gap}(N) < \lambda_{\max}$.